Project Work Book

(Guidelines and Log) **Course Code: 410248 & 410256 (2019 Course)**

Fourth Year of Computer Engineering

Year 2026 - 2027

Group/Project ID: | | | | |

Team Members:
1. Ajit Paraskar .....
2. Harshvardhan Patil .....
3. Prathamesh Kumbhar .....
4. Bikram Ghosh .....

Project Title : - - -

Savitribai Phule Pune University

Preamble

Project work is one of the most important components of the curriculum for an Engineering Graduate. Right from conceiving the idea to its materialization, is a journey that must be systematized, well defined and well documented to enjoy the full benefits of the efforts undertaken.

Every activity of the project development has its own importance. Team formation, conceiving the idea, preparing the hypothesis, reporting the progress and development to the guide(/mentor), Interactions, suggestions and improvements, relevant documentations in proper format, schedule plans and visit logs are some of the typical activities involved in project development.

Every institute is following their own best methods and techniques as per the guidelines and curriculum of the affiliated university. To bring uniformity for the project work there is a need to come together and prepare comprehensive guidelines and to standardize the process.

This project workbook will serve the purpose and facilitate the job of students, guide and project coordinator. This document will reflect accountability, punctuality, technical writing ability and work flow of the work undertaken.

This document will definitely support the work undertaken.

Dr. Varsha H. Patil
Coordinator, Board of Studies, Computer Engineering
Savitribai Phule Pune University
June 2018

General Instructions

1. Students should enter correct information in the work book.
2. Get all entries verified by respective project guide. No changes are to be made without project guide's permission.
3. Students should report to their respective guides as per the schedule and its log is to be maintained in the work book.
4. Follow all deadlines and submit all documents strictly as per prescribed formats.
5. The work book should be produced at the time of all discussions, presentations and examinations.
6. The work book must be submitted to project coordinator/ guide/ department / College after successful examination at the end of year.
7. All documents and reports are to be prepared in Latex only (All the formats specifications provided adheres to MS Word but consequently applicable to finalized project report published using Latex)
8. Submit hard as well as soft copy and maintain copy with each member.

This booklet is supportive document to rules and regulations enforced by affiliated university. This booklet provides recommendations, guidelines and is record of all related activities associated with project work. This booklet is provided with a genuine intent to bring uniformity, to systematize the project work and to keep audit of work undergone by team members.

Work Book Development Project

Project Institution	Department of Computer Engineering Matoshri College of Engineering and Research Centre, Nashik
Support & Guidance	Dr. Gajanan K. Kharate, Principal, Matoshri College of Engineering and Research Centre, Nashik
Concept and Design	Dr. Varsha. H. Patil BoS Coordinator Computer Engineering , SPPU, Pune Vice Principal, Matoshri College of Engineering and Research Centre, Nashik
Project coordinator	Mrs. Swati A. Bhavsar Assistant Professor, Matoshri College of Engineering and Research Centre, Nashik
Technical Committee Members	1. Dr. Goraksh V. Garje 2. Dr. Parikshit Mahalle 3. Mr. Niranjan L. Bhale 4. Dr. Sunil R. Dhore 5. Dr. Nuzhat Shaikh 6. Dr. Sudeep Thepade 7. Dr. Mangesh Bedekar 8. Ms. Swapnaja Hiray 9. Mr. Ranjeet Gawande
Date	5th January 2017
Version No. & Revision Date	3.0 (Revised on 10th June 2018)
Copyright (All rights reserved)	Registration Number: L-66962/2017 (Copyright document excludes Sr. No. 2-University Syllabus, Annxure- SE Code of Ethics and Professional Practices)

(For circulation at BoS Computer Engineering, Savitribai Phule Pune University only)

Savitribai Phule Pune University
Computer Engineering

Program Educational Objectives

PEO1. To prepare globally competent graduates having strong fundamentals, domain knowledge, updated with modern technology to provide the effective solutions for engineering problems.

PEO2. To prepare the graduates to work as a committed professional with strong professional ethics and values, sense of responsibilities, understanding of legal, safety, health, societal, cultural and environmental issues.

PEO3. To prepare committed and motivated graduates with research attitude, lifelong learning, investigative approach, and multidisciplinary thinking.

PEO4. To prepare the graduates with strong managerial and communication skills to work effectively as an individual as well as in teams.

Program Outcomes

Students are expected to know and be able -

PO1. To apply knowledge of mathematics, science, engineering fundamentals, problem solving skills, algorithmic analysis and mathematical modeling to the solution of complex engineering problems.

PO2. To analyze the problem by finding its domain and applying domain specific skills

PO3. To understand the design issues of the product/software and develop effective solutions with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

PO4. To find solutions of complex problems by conducting investigations applying suitable techniques.

PO5. To adapt the usage of modern tools and recent software.

PO6. To contribute towards the society by understanding the impact of Engineering on global aspect.

PO7. To understand environment issues and design a sustainable system.

PO8. To understand and follow professional ethics.

PO9. To function effectively as an individual and as member or leader in diverse teams and interdisciplinary settings.

PO10. To demonstrate effective communication at various levels.

PO11. To apply the knowledge of Computer Engineering for development of projects, and its finance and management.

PO12. To keep in touch with current technologies and inculcate the practice of lifelong learning.

Program Specific Outcomes (PSO)

A graduate of the Computer Engineering Program will demonstrate-

PSO1: Professional Skills-The ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, multimedia, web design, big data analytics, and networking for efficient design of computer-based systems of varying.

PSO2: Problem-Solving Skills- The ability to apply standard practices and strategies in software project development using open-ended programming environments to deliver a quality product for business success.

PSO3: Successful Career and Entrepreneurship- The ability to employ modern computer languages, environments, and platforms in creating innovative career paths to be an entrepreneur, and a zest for higher studies.

Table of Contents

Sr. No.	Description	Page No.
	Preamble	i
	General Instructions	ii
	Details of Work Book Development Project	Iii
	PEO, PO and PSO for Computer Engineering	iv
1.	Project work 1.1 Project Audit Committee (PAC) 1.2. Course Objectives 1.3 Course Outcomes 1.4 Mapping of Course Outcomes (CO) and Program Outcomes (PO) 1.5 Guidelines for Project Selection, Finalization and Guide Allotment 1.6 General Project Work Evaluation Parameters 1.7 Publication Guidelines 1.8 Intellectual Property Rights (IPR)	1
2.	University Syllabus	4
3.	Undertaking by Students	7
4.	Schedule of Project Work	8
5.	Project Review (Semester I)	
	6.1. Statement, Motivation, Objectives and Literature Review	
	6.2. Feasibility and Scope	
	6.3. Requirements Analysis	
	6.4. Design	
6.	Internal Evaluation Sheet (Semester I)	13
7.	Review Sheets (Semester II)	14
	8.1. Modeling	
	8.2. Coding and Implementation	
	8.3. Validation and Testing	
	8.4. Report Writing	
8.	Internal Evaluation Sheet (Semester II)	18
9.	Record of Contests Participation / Publications / IPR and Similar Event	19
10.	Rubrics A. Idea Inception B. Implementation C. Documentation D. Demonstration E. Contest Participation / Awards, Publications and IPR F. Environment & Ethics (solution to problems considering)	20

	Annexure	
	I. Format for Project Proposal	22
	II. Format for SRS	23
	III. Format for Partial Project Report (Semester I)	24
	IV. Format for Project Report (Semester II)	28
	V. Formatting guidelines for Reports	31
	VI. Format for Project Report Cover Pages	32
	VII. Format for Report Documentation Page	34
	VIII. Software Engineering code of ethics & professional practices	35
	IX. Copy of Project Proposal	37 onwards

1. Project Work

The word *project* comes from the Latin word *projectum* from the Latin verb *proicere*, "to throw something forwards" which in turn comes from *pro-*, which denotes something that precedes the action of the next part of the word in time (paralleling the Greek πρό) and *iacere*, "to throw". The word "project" thus actually originally meant" something that comes before anything else happens".

(Curtsey Ref- http://en.wikipedia.org/)

The intention of Project work is to conceive an idea and to implement it systematically by using knowledge derived during the course of education mainly to innovate or facilitate.

A group of Under Graduate students at Final Year will undertake project over academic year. Work involves study of feasibility of the project, planning of project, studying existing systems, tools available to implement the project and state of art software testing procedures and technology with use of case tools, design is to be implemented into a working model (software or hardware or both) with necessary software interface as an executable package.

1.1 Project Audit Committee (PAC):

It is recommended to form a departmental "Project Audit Committee" to monitor project activities comprising of Head, Project Coordinator, Industry Expert(s), External Expert(s), Department Academic Coordinator and few senior guides.

1.2. Course Objectives:

- To Apply the knowledge for solving realistic problem
- To develop problem solving ability
- To Organize, sustain and report on a substantial piece of team work over a period of several months
- To Evaluate alternative approaches, justify the use of selected tools and methods
- To Reflect upon the experience gained and lessons learned
- To Consider relevant social, ethical and legal issues,
- To find information for yourself from appropriate sources such as manuals, books, research journals and from other sources, and in turn increase analytical skills.
- To Work in TEAM and learn professionalism.

1.2 Course Outcomes:

Students are expected to know and be able to-

CO1. Knowledge Application & Independent Learning: Solve real life problems by applying knowledge and skills keeping eye on current technologies and inculcating the practice of lifelong learning

CO2. Problem Solving Skills: Analyze alternative approaches, apply and use most appropriate one for feasible solution exhibiting project management skills

CO3. Communication: Demonstrate effective communication at various levels and write precise reports and technical documents in a nutshell

CO4. Collaboration: Participate effectively in multi-disciplinary and heterogeneous teams exhibiting team work, Inter-personal relationships, conflict management and leadership quality

CO5. Ethics: Provide solution to problems considering social, safety, environmental, ethical and legal issues

1.3 Mapping of Course Outcomes (CO) of Project Work and Program Outcomes (PO):

The proper assessment of the COs and POs is one of the most important processes and it is to be done with precision and planning. It is recommended to assess the students continuously as they progress through the program. It is collectively one or more processes that define, collect, and prepare data to evaluate the achievement of Program Outcomes. Every COs are to be mapped to different POs based on their influence of COs on them. Sample mapping of PO and CO for project work is given in table 1.

Table 1: Mapping of CO and PO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√	√		√	√						√
CO2	√	√	√	√							√	
CO3											√	
CO4									√	√		
CO5						√	√	√				

1.4 Guidelines for Project Work Selection, Finalization and Guide Allotment:

Project is one of the significant contributory team works that has to be completed with distinct impression. It is necessary to explore the domain of interest / research/ thirst area/ society needs. In Toto one cannot figuratively define best project but still there are certain parameters on which we can gauge the quality of project work done. It will be better suited to go for well-defined and relatively safe projects that provide scope for demonstrating proficiency with a low risk of failure especially at Under Graduate level.

Process in General:

1. Project teams and their areas of interest is to be registered with project Coordinator preferably in second semester of third year.
2. Students are provided with list of guides & their domain of expertise, list of earlier three years projects, constitution of PAC and copy of logbook giving all guidelines.

3. Considering registered teams area of interest/domain and expertise of guide, the Project coordinator in consultation with PAC tentatively allots Project guides.
4. Team may come up with sponsored project (Title suggetion and associated guidance by external institute/Company).
5. Teams in consultation with guide prepare project proposal(s)
6. Project Proposal must include project title, group members, sponsorship details (if any), detailed problem definition, area, Type of Project [Sponsored/Non Sponsored, AND viz- 1. Framework, 2. System as - Application/ Systems Software with or without Hardware 3. Research, 4. Survey], abstract, details of existing similar systems if any, scope of the project and software-hardware requirements. [Sponsorship details include name of sponsoring authority, address, name of guide, sponsorship terms and conditions and respective documents certifying the same from authorities].
7. A Panel of experts will approve the project group and title. Discussion / presentation may be arranged covering topics listed in the proposal.
8. Once project titles are finalized by PAC, guides are reallocated/ changed, if required.
9. It is recommended to seek guidance from PG students and/or alumini and assistance from third year students.
10. It is recommended to maintain record of all meetings, discussions, suggestions, contributions and roles played by each member of the team.

Dos and Don'ts:

- Project work is expected to involve a combination of study (literature study/ line of investigation), and methodical implementation.
- Instead of fancied and driven behind the gaudy and ostentatious ideas, utility needs to be emphasized. It is also acceptable to identify the discrepancies/ flaws an existing system and work accordingly to rectify or improve.
- It is irrational to select the IDE and the software/ tools before the idea is not yet finalized.
- Identify domain, feasibility and usability of work.
- Understand the way project will materialize and progress is of at most importance.

1.5 General Project Evaluation Parameters:

Project work is to be evaluated jointly by both Internal and External examiners, unanimously agreeing upon the following parameters amongst many others.

1. Problem definition and scope of the project.
2. Thorough literature survey done.
3. Exhaustive and rational requirement analysis.
4. Appropriate software engineering approach followed.
5. Use of project management tools.
6. Comprehensive implementation

7. Optimization considerations(memory, time, resources, costing).
8. Use of parallel/multi-core, embedded, distributed computing approach.
9. Thorough testing of all modules and integration of modules done.
10. Project presentation and demonstration.
11. User interface, ease of use, usability and GUI.
12. Understanding individual capacity, role and involvement in the project.
13. Team work (roles defined, distribution of work, intra-team communication and togetherness).
14. Participation in various contests, publications and IPR.
15. Presentation of work in the form of project report(s). Documents /manuals - project report, quick reference, system, installation guide etc
16. Outcomes / usability/ commercial value/ product conversion of work
17. Consideration of social, safety, environmental, ethical and legal issues

1.8. Publications Guidelines:

The work undertaken is to be appreciated and recognized by the significant publications and/or IPR. The quality of the publications reflects the efforts and recognition of the work. So, it is highly recommended to publish work in consultation with the guide in referred national and international Journals of repute, with high Impact Factor and also in recognized conferences. There are some journals operating in different regions which use 'International' word, but in true sense are not International. Refer

1. http://www.fi.dk/viden-og-politik/tal-og-analyser/den-bibliometriske-forskningsindikator/autoritetslister-for-tidsskrifterog-forlag/bfi-publishers-2011.pdf
2. http://www.fi.dk/viden-og-politik/tal-og-analyser/den-bibliometriske-ingsindikator/autoritetslister-for-tidsskrifter-og-forlag/Autoritetslisten%20for%20tidsskrifter%202011%20-20med%20niveauer.pdf

1.9. IPR Guidelines:

The first legislation in India for protection of Industrial Designs was The Patents & Designs Protection Act, 1872. It supplemented the 1859 Act passed by Governor General of India for granting exclusive privileges to inventors and added protection for Industrial Design. The 1872 Act included the term —any new and original pattern or design, or the application of such pattern or design to any substance or article of manufacture'.

Hence it is recommended that students should know about Copyright and Patents.

Refer-

1. http://www.ipindia.nic.in
2. http://www.ipindia.nic.in/writereaddata/Portal/IPOGuidelinesManuals/1_30_1_manual-designs-practice-and-procedure.pdf

2. University Syllabus (semester I)

http://collegecirculars.unipune.ac.in/sites/documents/Syllabus%202018/Forms/AllItems.aspx?InitialTabId=Ribbon%2EDocument&VisibilityContext=WSSTabPersistence)

Project Work Stage I

Course Objectives:

- To Apply the knowledge for solving realistic problem
- To develop problem solving ability
- To Organize, sustain and report on a substantial piece of team work over a period of several months
- To Evaluate alternative approaches, and justify the use of selected tools and methods,
- To Reflect upon the experience gained and lessons learned,
- To Consider relevant social, ethical and legal issues,
- To find information for yourself from appropriate sources such as manuals, books, research journals and from other sources, and in turn increase analytical skills.
- To Work in TEAM and learn professionalism

Course Outcomes:

On completion of the course, student will be able to-

- Solve real life problems by applying knowledge.
- Analyze alternative approaches, apply and use most appropriate one for feasible solution.
- Write precise reports and technical documents in a nutshell.
- Participate effectively in multi-disciplinary and heterogeneous teams exhibiting team work, Inter-personal relationships, conflict management and leadership quality

Guidelines-

Project work Stage – I is an integral part of the Project work. In this, the student shall complete the partial work of the Project which will consist of problem statement, literature review, design, scheme of implementation (Mathematical Model/SRS/UML/ERD/block diagram/ PERT chart, etc.) and Layout & Design of the Set-up. The student is expected to complete the project up to the design phase. As a part of the progress report of Dissertation work Stage-I, the candidate shall deliver a presentation on the advancement in Technology pertaining to the selected project topic. The student shall submit the duly certified progress report of Project work Stage-I in standard format for satisfactory completion of the work by the concerned guide and head of the Department/Institute.

The examiner will be assessed by a panel of examiners of which one is necessarily an external examiner. The assessment will be broadly based on work undergone, content delivery, presentation skills, documentation and report.

Semester II

Project Work Stage II

`

Course Objectives:

- To follow SDLC meticulously and meet the objectives of proposed work
- To test rigorously before deployment of system
- To validate the work undertaken
- To consolidate the work as furnished report.

Course Outcomes:

On completion of the course, student will be able to–

- Show evidence of independent investigation
- Critically analyze the results and their interpretation.
- Report and present the original results in an orderly way and placing the open questions in the right perspective.
- Link techniques and results from literature as well as actual research and future research lines with the research.
- Appreciate practical implications and constraints of the specialist subject

Guidelines-

In Project Work Stage–II, the student shall complete the remaining project work which consists of Selection of Technology and Tools, Installations, UML implementations, testing, Results, performance discussions using data tables per parameter considered for the improvement with existing/known algorithms/systems and comparative analysis and validation of results and conclusions. The student shall prepare and submit the report of Project work in standard format for satisfactory completion of the work that is the duly certified by the concerned guide and head of the Department/Institute

3. Undertaking by Students

Dr.D.Y.Patil institute of Technology

Pune-18

With reference to circular (ref-project/2009/3369) regarding malpractices in project work from DTE, Pune following undertaking is to be submitted.

UNDERTAKING BY STUDENT

We, the students of B.E. Computer hereby assure that we will follow all the rules and regulations related to project activity for the academic year 2026-2027. The Project titled-

"Cortex: An Agentic Retrieval-Augmented Generation System over Live Work Data with Explainable Memory and Consent-Gated Actions"

will be fully designed/ developed by us and every part of the project will be original work and will not be copied/ purchased from any source.

Name of the student	Signature
1. Ajit Paraskar	
2. Harshvardhan Patil	
3. Prathamesh Kumbhar	
4. Bikram Ghosh	

4. Schedule of Project Work

Semester I

Sr. No.	Activity Scheduled	Date (Tentative)
1.	Registration of Project Teams	Third Year Semester II/ Mid of June
2.	Submission of Project Proposal	Last Week of June
3.	Project presentations	First week of July
4.	Finalization of projects & allotment of guide	Second week of July
5.	Submission of final Proposal	Third week of July
6.	Review meeting/ presentation for progress of project work- I	Last week of July
7.	Review meeting/ presentation for progress of project work- II	Third week of August
8.	Review meeting/ presentation for progress of project work- III	Second week of Sept
9.	Review meeting/ presentation for progress of project work- IV	Last week of Sept
10.	Submission of partial project report	1st Week Oct
11.	Project work (Stage I) Examination	As per SPPU Notification

Semester II

Sr. No.	Activity Scheduled	Date(Tentative)
1.	Review meeting/ presentation for progress of project work -V	Second week of Jan
2.	Review meeting/ presentation for progress of project work -VI	Second week of Feb
3.	Review meeting/ presentation for progress of project work- VII	Last week of March
4.	Submission of final project report and Project Work book to the project Coordinator	First week of April
5.	Project Examination	As per SPPU Notification

5. Project Review (Semester I)

The group members are expected to present their work undertaken during the semester. Journey of development has to be rationally presented with thorough literature survey in review meeting.

5.1 Project Review-I: Problem Statement, Motivation, objectives and Literature Review

Student is expected to deliver presentation covering Problem Statement, Motivation, objectives and Literature Review.

Sr. No.	Question	Date	Remark / Grade	Sign of Guide
1)	Do Research gap identified lead to find motivation of project? **Ans:** *Yes. The survey shows that RAG methods use static indexes, the strongest search policies (GRASP, Self-RAG, Search-R1) need training, claims are not audited per sentence, conflicts between sources are resolved silently, agent memory is opaque and tool-using agents act without per-action consent. Closing these gaps is the motivation of Cortex.*			
2)	Does the statement give clear identification about what your project will accomplish? **Ans:** *Yes. Cortex will answer questions over a user's live work tools with cited and verified answers, report conflicts and missing information, keep an explainable memory and perform actions only after explicit approval, all on free-tier language models.*			
3)	Is the statement short and concise? **Ans:** *Yes. In short: a free-to-run agentic RAG system that answers questions over live work data with cited, verified answers and acts only with the user's consent. The full statement is one sentence (Annexure IX).*			
4)	Do similar type of methodology / model exists? **Ans:** *Yes. Related methods are RAG, ReAct, IRCoT, Self-Ask, Self-RAG, CRAG, Adaptive-RAG and the base paper GRASP; related products include Supermemory, MemoryFirst and Augmas. Cortex differs by being training-free, working over live tools, verifying every claim and gating every action.*			
5)	Is the studied literature sufficient to decide scope of the project? **Ans:** *Yes. More than twenty works on retrieval-augmented generation, agentic retrieval, attribution, agent memory, tool use (MCP) and long contexts were studied and compared (comparison table in the report).*			
6)	Are the objectives clearly and unambiguously listed? **Ans:** *Yes. Eight objectives are listed (Annexure IX, Part A), e.g. cite and verify every claim, never execute an action without approval, keep LLM calls per question bounded.*			

7)	Can a person who is not familiar with the project understand scope of the project by reading the project problem statement? **Ans:** *Yes. The scope states what the product is (web app, reusable packages, memory service) and what it is not (no data warehouse, no model training, no autonomous writes), with numeric input bounds.*			
8)	Are project objectives of study (what product, process, resource etc.) clearly defined? **Ans:** *Yes. Product: web application with agent, connector, model-router packages and a memory service. Process: bounded plan-then-execute pipeline. Resources: free-tier LLM, connector and database services.*			
9)	Are the objectives set helpful to achieve goal of the project? **Ans:** *Yes. Each objective maps to a measurable outcome: groundedness score, citation recall, number of actions executed without consent (must be 0) and LLM calls per question.*			
10)	Does the project contribute to our society by any means? **Ans:** *Yes. It reduces time spent searching scattered information, shows evidence for every claim (reducing misinformation), protects users' accounts through consent, and is free to run for students and individual developers.*			
Remark and Suggestions:				

Name and Sign of Reviewers:

1.

2.

5.2 Project Review-II: Feasibility and Scope

Student is expected to deliver presentation covering Feasibility and Scope

Sr. No.	Question	Date	Remark / Grade	Sign of Guide
1)	Is the project's view point understood? **Ans:** *Yes. For the user it is a research assistant over their own tools that shows its evidence; for the developer it is a bounded pipeline with predictable cost.*			
2)	Is the project goal statement in alignment with the sponsoring organization's business goal and mission? **Ans:** *Not applicable: this is an internal, non-sponsored project. It is aligned with the department's goals for project work in AI and information retrieval.*			
3)	Who is the project's end user? **Ans:** *Developers, productivity-focused professionals and undergraduate students whose work is spread across Gmail, Google Calendar, Google Drive, Notion, Slack and GitHub.*			
4)	What is the projected cost of producing a product? **Ans:** *Operating cost is INR 0 (all services on free tiers). Basic COCOMO on about 6.1 KLOC gives about 16.0 person-months of effort.*			
5)	Is project achievable in specified (Time, Cost Budget)? **Ans:** *Yes. COCOMO gives a duration of about 7.2 months with about 2.2 persons on average; the four-member team has the academic year, and the monetary cost is zero.*			
6)	Are the requirements within the scope of the project? **Ans:** *Yes. Deferred vision items (time-travel queries, editable plan, Cortex as an MCP server) are explicitly kept out of scope.*			
7)	Is the scope properly defined? **Ans:** *Yes, with explicit bounds: question <= 2000 characters, <= 5 sub-questions, <= 18 searches, <= 4 LLM calls (Deep), 90 s per run.*			
8)	Does the problem statement clearly define scope of the project? **Ans:** *Yes. The problem statement fixes the inputs (question over connected tools), outputs (cited, verified answer, conflicts, proposals) and constraints (bounded calls, consent, free tiers).*			
9)	Do the project requirements fit into available software and hardware? **Ans:** *Yes. Development runs on ordinary laptops (8 GB RAM) with free and open-source software;*			

	Oracle AI Database is optional and a SQLite fallback is used.			
10)	Whether the milestones are stated completely and project timeline is given? **Ans:** *Yes. Phases (survey, SRS, design, implementation, testing and evaluation, documentation) and a Gantt chart are in the project report; the dates there are planned and are to be confirmed with the guide.*			
11)	Whether risks like technical risks, Operational risks, schedule risks, business risks are identified correctly or not? **Ans:** *Yes. Six risks are identified: free-tier limits, connector API changes, hallucinated claims, unintended actions, unavailable Oracle backend, and exposure of private data.*			
12)	Whether Risk prioritization is done properly and any back up plan is decided? **Ans:** *Yes. Risks are ranked by probability and impact. Back-up plans: provider failover and caching; tool names stored as data; claim verification; single approval endpoint; SQLite fallback; no storage of source data.*			
Remark and Suggestions:				

Name and Sign of Reviewers:

1.

2.

5.3 Project Review-III: Requirement Analysis

Student is expected to deliver presentation covering Requirement Analysis

Sr. No.	Question	Date	Remark / Grade	Sign of Guide
1)	Is information domain analysis complete, consistent and accurate? **Ans:** *Yes. Data objects (Evidence, Plan, SubQuestion, Claim, Conflict, ActionProposal, MemoryItem, SourceFailure, AgentEvent) and their relationships are identified and shown in an ERD.*			
2)	Is problem statement categorized in identified area and targeted towards specific area there in? **Ans:** *Yes. Information retrieval and NLP (ACM H.3.3, H.3.4, I.2.7, I.2.11), targeted at multi-source question answering with verification.*			
3)	Is external and internal interfacing properly defined? **Ans:** *Yes. Browser-server REST + server-sent events, memory service HTTP/JSON API, OpenAI-compatible LLM APIs, Composio SDK / MCP and the Firecrawl API are specified.*			
4)	Are requirements consistent with schedule, resources and budget? **Ans:** *Yes. All services are on free tiers and every run is bounded in LLM calls, tool calls and time.*			
5)	Are all requirements traceable to system level? **Ans:** *Yes. Each functional requirement maps to a component, e.g. ask -> orchestrator, approve -> ActionStore via /api/actions/{id}, memory -> memory service.*			
6)	What is needed to make the product? **Ans:** *Node.js 24+, Python 3.11, Next.js/React, free-tier API keys (Groq, Gemini, Cerebras, OpenRouter, Puter), Composio and Firecrawl accounts, and a laptop with Internet access.*			
7)	Is there a demand for the product? **Ans:** *Yes. Interaction workers spend about 28% of the week on e-mail and nearly 20% searching for information (McKinsey Global Institute); several commercial assistants address this market, mostly paid.*			
8)	Is identification of stakeholders done properly? **Ans:** *Yes. End user, operator, internal guide/department, connected-application providers and LLM providers.*			
9)	Whether all requirements are captured and documented in line with scope? **Ans:** *Yes. They are documented in the SRS (Annexure IX, Part B) and in the project report.*			

10)	Whether all type of analysis classes are identified? **Ans:** *Yes. Boundary: web pages and API routes; control: orchestrator, verifier, ActionStore, LLM router; entity: Evidence, Claim, MemoryItem, ActionProposal.*			
11)	Whether the Acceptance criteria is decided **Ans:** *Yes. Every factual sentence cites existing evidence; groundedness and conflicts are reported; no action executes without approval; failed sources are named with their impact; all automated tests pass.*			
12)	Is SRS document as per IEEE format complete and correct? **Ans:** *Yes. The SRS follows the IEEE-style structure of the department template: purpose and scope, usage scenarios, data model, functional model, non-functional requirements, constraints and interfaces.*			
Remark and Suggestions:				

Name and Sign of Reviewers:
1.

2.

5.4 Project Review-IV: Design

Student is expected to deliver presentation covering Design

Sr. No.	Question	Date	Remark / Grade	Sign of Guide
1)	Are requirements reflected in the system architecture? **Ans:** *Yes. The layered architecture (browser, web server, agent, model router, connectors, memory service) has one component for each functional requirement.*			
2)	Does the design support both project (product) and project goals? **Ans:** *Yes. Bounded calls keep the cost at zero, verification supports trust, and the single approval endpoint guarantees consent.*			
3)	Does the design address all the issues from the requirement? **Ans:** *Yes, including failure handling: failed sources become source_error events, and timeouts are set per connector call, model request and run.*			
4)	Is effective modularity achieved and modules are functionally independent? **Ans:** *Yes. Separate packages (agent, connectors, llm, shared) and a separate memory service; dependencies are injected through one AgentDeps object.*			
5)	Are structural diagrams (class, Object, etc) well defined? **Ans:** *Yes. A class diagram in two parts (core and connectors), plus package and deployment views.*			
6)	Are all class associations clearly defined and understood?(Is it cleat which classes provide which services)? **Ans:** *Yes. AgentDeps composes LlmRouter, ConnectorRegistry, MemoryBackend and ActionStore; ConnectorRegistry holds one or more Connectors.*			
7)	Are the classes in the class diagram clear? (What they represent in the architecture design document?) **Ans:** *Yes. Each class corresponds to a module in the code, e.g. LlmRouter in packages/llm and ActionStore in packages/agent.*			
8)	Is inheritance appropriately used? **Ans:** *Interface realisation is used instead of deep inheritance: five connector classes implement Connector, and MemoryClient implements MemoryBackend.*			
9)	Are the multiplicities in the use case diagram depicted in the class diagram? **Ans:** *Yes. A run has one plan, a plan has 1..5 sub-questions, and a run has 0..3 action proposals and 0..2 memory writes.*			
10)	Are all objects used in sequence diagram? **Ans:** *Yes. User, Web UI, Agent, ActionStore and Connector appear in the sequence diagram and in the class model.*			
11)	Are the symbols used in all diagrams corresponding to UML standards? **Ans:** *Yes. Standard UML notation is used in all diagrams.*			

12)	Are behavioral diagrams (use case, sequence, activity, etc.) well defined and understood? **Ans:** *Yes. Use case, activity, state and sequence diagrams are drawn.*			
13)	Does each case have clearly defined actors and input/ output? **Ans:** *Yes. The use-case table lists the actors, a description and the assumptions for each use case.*			
14)	Does the sequence diagram match with class diagram? **Ans:** *Yes. ActionStore.propose/decide and Connector.execute appear in both diagrams.*			
15)	Is aggregation/ containment (used) clearly defined and understood? **Ans:** *Yes. Composition from AgentDeps to its parts and aggregation of connectors in ConnectorRegistry.*			
16)	Whether State charts are capturing system's dynamic behavior correctly? **Ans:** *Yes. The state diagram of an action proposal covers Pending, Approved, Rejected, Expired, Executed, Failed and Discarded.*			
17)	Related to procedural thinking whether DFDs and CFDs along with transaction and transformation flow are done correctly ? **Ans:** *Yes. Level 0 and Level 1 DFDs are drawn; Level 1 shows the transformation flow through nine processes, with the approval endpoint as the transaction centre.*			
Remark and Suggestions:				

Name and Sign of Reviewers:

1.

2.

6. Internal Evaluation Sheet (Semester I)

Sr. No.	Names of Team Members	Problem Statement / Motivation / Objectives / Scope/ Feasibility Requirement (05)	Literature Survey (05)	Requirement Analysis (05), Modeling & Designing (10)	Planning & Prototyping (05)	Presentation & Question - Answer (10)	Partial Project Report (10)	Total (50)
1.								
2.								
3.								
4.								

Name and Signature of Evaluation Committee:

-

1. **Prof.**
2. **Prof.**

Examiners Feedback and Suggestions:

Signature of Guide **PAC** **Signature of Head**
[Name of Guide**]** [Name of HoD]
Head of Department

7. Project Review: (Semester II)

The group members are expected to present their work undertaken during the semester. Journey of development has to be rationally presented.

7.1 Project Review-I: Modeling (Model Refinement and Algorithm development)

Student is expected to deliver presentation covering Modeling

Sr. No.	Question	Date	Remark/ Grade	Sign of Guide
1)	Which software Development Process model is used? (Water fall, Incremental, RAD) How? (? at this level?)			
2)	Are data objects, their attributes and relationships clearly identified? (All constraints fro SRS are captured or not?)			
3)	Have the objects and respective classes and their responsibilities?			
4)	Have you analyzed the requirements been analyzed and represented into respective models?			
5)	Have the different system states been differentiated and depicted them in the form of state transition diagram?			
6)	Does the mathematical model clearly imply design of the project?			
7)	Does the mathematical model clearly states goal of project?			
8)	Is the interface between the modules properly identified?			
9)	Are any functional dependencies identified and described?			
10)	Which architectural model does the system support?			
11)	Whether Deployment diagram is in line with selected architecture?			
12)	Whether all components are designed properly and represented in component diagram?			
13)	Whether NP-completeness of algorithms is checked?			
Remark and Suggestions:				

Name and Sign of Reviewers:

1.

2.

7.2 Project Review-II: Coding / Implementation

Student is expected to deliver presentation covering Coding / Implementation

Sr. No.	Question	Date	Remark/ Grade	Sign of Guide
1)	Does the code completely and correctly implement the design?			
2)	Does the code comply with the coding standard?			
3)	Is the code well structured, consistent in style, and consistently formatted?			
4)	Are all functions in the design coded?			
5)	Does the code make use of object oriented concepts?			
6)	Does the code support granularity?			
7)	Is the language used for coding correctly chosen as per the project need?			
8)	If any off-the-shelf components are used, Have you understood the functionalities of using it?			
9)	Are all comments consistent with the code?			
10)	Whether code optimization is done properly?(By using language features)			
Remark and Suggestions:				

Name and Sign of Reviewers:

1.

2.

7.3 Project Review-III: Validation and Testing

Student is expected to deliver presentation covering Validation and Testing

Sr. No.	Question	Date	Remark/ Grade	Sign of Guide
1)	Has alpha testing been done?			
2)	Has beta testing been done?			
3)	Have been validated the requirements, design and code as per standard?			
4)	Has GUI testing of project been performed? How?			
5)	Does the system comply with basic usability norms?			
6)	Has the code been tested using standard datasets available in your area of project?			
7)	Has the code been tested in real time environment?			
8)	After integration of all components whether total performance of system is checked?			
9)	Whether repository of all components along with versions is documented?			
10)	Have social, safety, environmental, ethical and legal issues been considered while providing solution to problem?			
Remark and Suggestions:				

Name and Sign of Reviewers:

1.

2.

7.4 Project Review-III: Report Writing

Student is expected to deliver presentation covering Report Writing

Sr. No.	Question	Date	Remark/ Grade	Sign of Guide
1)	Is the report written as per the prescribed format?			
2)	Is the report timely prepared?			
3)	Is the report properly organized, spelled, grammatically correct?			
4)	Is the report plagiarism free?			
5)	Is the report precise and written to the point?			
6)	Does the report contain complete results and comparative graphs?			
7)	Are all figures and tables properly numbered and labeled?			
8)	Are all figures and tables properly cited?			
9)	Weather references are properly cited?			
Remark and Suggestions:				

Name and Sign of Reviewers:

1.

2.

8. Internal Evaluation Sheet (Semester II)

Sr. No.	Names of Team Members	Modeling (10)	Coding and Implementation (40)	Testing (10)	Understanding, Individual Involvement / Contribution in the project (10)	Team Work (10)	Demonstration cum Presentation (10)	Documents & Report (10)	Total (100)
1.									
2.									
3.									
4.									

Name and Signature of Evaluation Committee:

1. **Prof.**
2. **Prof.**

Examiners Feedback and Suggestions:

Signature of Guide
[Name of Guide**]**

PAC

Signature of Head
[Name of HoD]
Head of Department

9. Contest Participation Details

9.1. Participation in Project Competition

Sr. No.	Name and Place of Project Competition and Exhibition	Date	Certificates / prizes won, if any
1.			
2.			
3.			
4.			

Attach attested copy of certificate(s)

9.2. Paper Publication/ Presentation/IPR

Sr. No.	Name of Organizer	Date	Certificates/ Prizes won, if any
1.			
2.			
3.			
4.			

Attach attested copy of certificate(s)

10. Rubrics

A. Idea Inception

Grade (Grade Point)	Excellent (10-9)	Very Good (6-8)	Fair (3-5)	Poor (1-2)
Parameter				
Problem Definition and Scope of the Project				
Literature Survey				
Software Engineering Approach				
Requirement Analysis				

B. Implementation

Grade (Grade Point)	Excellent (10-9)	Very Good (6-8)	Fair (3-5)	Poor (1-2)
Parameter				
Implementation- Design, platform, coding,				
Optimization considerations(Memory, time, Resources, Costing)				
Thorough Testing of all modules				
Integration of modules and project as whole				

C. Documentation

Grade (Grade Point)	Excellent (10-9)	Very Good (6-8)	Fair (3-5)	Poor (1-2)
Parameter				
Proposal				
Project Report				
Quick references				
System manual				
Installation Guide				
Work Book				

D. Demonstration

Grade (Grade Point)	Excellent (10-9)	Very Good (6-8)	Fair (3-5)	Poor (1-2)
Parameter				
Project Presentation and Demonstration(User Interface, ease of use, usability)				
Understanding individual capacity & involvement in the project				
Team Work (Distribution of work, intra-team communication and togetherness)				
Outcomes / Usability				

E. Contest Participation / Awards, Publications and IPR

Grade (Grade Point)	Excellent (10-9)	Very Good (6-8)	Fair (3-5)	Poor (1-2)
Parameter				
Participation in various contests				
Appreciation and Awards				
Publications				
Copyright				
Patent				
Commercial value /product conversion of Work				

F. Environment & Ethics (solution to problems considering)

Grade (Grade Point)	Excellent (10-9)	Very Good (6-8)	Fair (3-5)	Poor (1-2)
Parameter				
social				
safety				
environmental				
ethical				
Legal issues				

Annexure I: Format for Final Proposal

Title Page

- Project Group ID
- Title of the project
- Domain such as Databases, Data Mining, Image processing, WSN, Web technology etc.
- Team Member (PRN, Roll No, Name, Role/Task assigned, email, mobile & sign)
- Sponsorship details, if any (Name, External Guide name and Designation with Signature, e- Mail ID)
- Internal Guide(with signature of approval)
- Type of Project
- Mentor(PG Student/Alumni/other)
- TE students associated with:(Roll No, Name)

Inner Pages:

- Keywords (ACM Keywords)
- Problem Definition
- Abstract
- Process Diagram, System architecture, List of modules, and functionalities
- Literature survey (Study, Current market survey: similar systems/products available, their pros and cons.
- Objectives
- Scope of the project
- Software and hardware requirements
- Expected Outcomes
- Probable date of completion
- References (This should include the list of books, magazines, research papers, web links etc. referred by the students)

Annexure II: System Requirement Specification

Software requirement Specification is a detailed write-up indicating the requirements that the project demands. It contains actual detailed problem definition. The definition should include all that is to be done and is to be developed in the final software and / or Hardware (product) that has to be generated form the years work (User's point of view). The entries under this section are to be categorized as,

1. Necessary functions,
2. Desirable functions, and others

Requirement may not be final and provision should be available to add features dynamically without affecting the actual flow and design of the document. Modified requirements (after doing feasibility study) are to be prepared under all the 3 categories
listed above form the developer's point of view. The requirements listed herein should be feasible technically form the software/ Hardware point of view.

It should include following important requirements.

1. Detailed Problem Definition

2. External Interface Requirements

- User interfaces
- Hardware Interfaces
- Software Interfaces
- Communication Interfaces

3. System Features

- Feature 1
- Feature 2 etc.

4. Other Non- functional requirements.

- Performance requirements
- Safety requirements
- Software Quality attributes

Annexure III Partial Project Report (Semester I)

A preliminary report of project work (Partial Project Report) is to be prepared as per the guideline given below using **Latex** and is to be submitted at the end of semester I.

Title Page

First page containing Name, Topic Name, Guide Name, Year, Branch, and College Name etc.

Certificate

- Certificate
- Project approval sheet
- Certificate by the sponsoring authority, if any.

Acknowledgements (if any)

Thanking any person / staff member / friend if to be done so.

Abstract

A minimum of 100 words briefing the topic in consideration.

Keywords

A minimum of 5 and maximum of 10

Introduction

Introduction includes briefing of the details to follow, details of project work, objectives, scope of the project, motivation and organization of report.

Literature Survey

The purpose of the literature survey is to identify information relevant to project work and the potential and known impacts of it within the project area. This section should include a comprehensive report of current market survey done with respect to problem. Include study of similar systems available, if any along with their pros and cons. identify those areas where there is an absence or scarcity.

Design Details (Phase I to IV)

Phase I: Requirements Analysis

The Group is to submit a detailed write – up indicating the requirements that the project demands, viz.

- Actual detailed problem definition.
- The definition is to include all that is to be done and is to be put up in the final software and / or
- Hardware (product) that is to be generated from the years work (User's point of view).

Requirement may not be final and provision should be available to add features dynamically without affecting the actual flow and design of the document.

Modified Requirements (After doing feasibility study) are to be prepared under all the 3 categories listed above from the developer's point of view. The requirement listed herein should be feasible technically from the software / Hardware point of view.

Follow the standard format of SRS.

Phase II: Analysis Phase

The group (based on Phase I) is to suggest the paradigm followed by them in the project. The paradigm should be justifiable from Phase I. The various stages and work to be completed under them is to be indicted in detail.

Phase III: Design Phase

ERDs (Optional, decide in consultation with guide)

- The group is to draw the ERD (Entity Relationship Diagram) for the project. (This should be justifiable with regard to Phase I & II)
- The ERD after getting evaluated (by dry running) is to be analyzed for incompleteness from any point of view.
- The ERD thus validated should be made fair in a presentable fashion.
- This ERD is to be included in the Report.

IF

The project group is to follow an "Object Oriented" Approach for their Project.

THEN

- The group should all UML (Unified Modeling Language) diagrams for the project.
- These diagrams are to be refined in every aspect for this report(as per requirements finalized in phase I)
- Proper notations are to be used in all the figures drawn.
- Proper Color-coding if required is to be used.
- Extensions to diagrams / customizations may be done and represented (if the project demands it)

ELSE (groups following Structured Approach)

- The group should draw the DFD-s (Data Flow Diagrams) for the Project. (These should be justifiable with respect to Phase I, II and the ERD)
- DFD Level 0, Level 1, Level 2 should be drawn in an evolutionary fashion (No entries to appear in Level 2 unless they are in Level 1, which in turn are in Level 0)
- The DFD's are to be validated and made final in a presentable fashion.
- Proper Color- coding is expected
- Extensions to DFD-s may be represented (if the project demands it)

Phase – IV: Planning Phase

- The group should finalize the Front End/ Back End required for the project as per the demands of the project(Software and / or hardware)
- The Front End/ Back End should be justifiable depending on the complexity of the project.
- The structure of the database should be finalized depending on the complexity of the project.
- Any Normalization required on the database is done so as to ensure correctness for the future phase.
- Coding Language / Methodology should be finalized/
- Time requirement to be finalized and indicated
- Actual project plan including major milestones should be decide and finalized
- Rough estimates of lines of code / functions / routines to be made.
- Rough estimates of lines of code / Objects / Classes to be made (for Groups following OO Paradigm)

- Software Reuse /Re – Engineering possibilities are to be expected and indicated
- Software and Hardware requirement.
- Probable date of completion.
- Scope of the project.

Phase – V: Prototyping

- A prototype is expected which basically includes all the MAJOR features in the project.
- The GUI/ Front end should be prepared.
- The structure of the database / back end (if any) to be indicated.
- The prototype is built basically to give a feel of the actual software and / or hardware (Product) that is expected
  Major routines / Functions are expected.

Conclusions

Write conclusions drawn from the work done with atleast 50 words.

References

List out Books, Magazines, Thesis, Journals, Web links etc referred in IEEE format

Plagiarism Check Report

Annexure IV: Format for Project Report (Semester II)

A report of project work has to be prepared as per the guidelines given below using Latex and should be submitted at the end of semester II along with CD containing (copy of Partial Project Report, Final Project Report along with .tex files, Power point presentation, copy of base paper and reference papers, executable Project Code, supportive software platform for the project execution).

- **First page as per standard college reports**
  First page containing Name, Topic Name, Guide Name, Year, Branch, and College Name etc. (see format displayed herewith)
- **Certificate** (will be provided by college)
  Dissertation approval sheet (see format displayed herewith) Also attach certificate certifying the project work done approved by the sponsoring authority, if any.
- **Abstract**
  A minimum of 100 words briefing the topic in consideration.
- **Keywords**
  A minimum of 5 and maximum of 10
- **Index**
  Details of various Topics, Sub-Topics, with Page No. Figure Index, giving details of page number, figure number and figure caption Table Index, giving details of page number, table number and table caption (If any) Index of Pseudo-code / Sample code (If any)

I) Introduction

Minimum of 200 words, giving some briefing of the details to follow.

· Detailed problem definition

· Justification of problem

· Need for the new system

· Advances/additions/updating the previous system

· Presently available systems for the same

· Purpose of your system

· Organization of the report

This section should be relevant to the Literature Survey done and reported in the partial project report. The purpose of the literature survey is to identify information relevant to project work and the potential and known impacts of it within the project area. This section should include a comprehensive report of current market survey done with respect to problem. Include study of similar systems available, if any along with their pros and cons. identify those areas where there is an absence or scarcity.

II) Analysis

· Project plan
· Requirement analysis
· Team structure

The Group has to submit a detailed write-up indicating the requirements that the project demands-

Actual detailed problem definition
The definition should include all that has to be done and developed in the final software and / or Hardware (product) that will be generated from the years work (User's point of view).

The entries under this section are to be categorized as-
1. Necessary functions,
2. Desirable functions,
3. Others

Requirement may not be final and provision should be available to add features dynamically without affecting the actual flow and design of the document. Modified Requirements (after doing feasibility study) are to be prepared under all the 3 categories listed above from the developer's point of view. The requirements listed herein should be feasible technically from the Software / Hardware point of view. The new list should be categorized in the 3 categories listed above. (Follow the IEEE format of SRS)
The group is to suggest the Paradigm followed by them in the Project. The Paradigm should be justifiable from Phase I. The various stages and work to be completed under them has to be indicted in detail.

III) Design

· Software Requirement Specification(SRS) format is as given below.
· Risk assessment
· Brief discussion on Project plan submitted in semester I including major milestones and the work done as per it.

IV) Modeling

· UML diagrams (all 9)
· ERD & Normalization (NF) for database (if any)

ERDs (Optional, decide in consultation with guide)

· The group should draw the ERD (Entity Relationship Diagram) for the Project. (This should be justifiable with regard to Phase I & II)
· The ERD after getting evaluated (by dry running) should be analyzed for incompleteness from any point of view
· The ERD thus validated should be made fair in a presentable fashion
· This ERD is to be included in the Report

IF
The project group is to follow an "Object Oriented "Approach for their Project.
THEN

· The group should prepare all UML (Unified Modeling Language) diagrams for the project
· These diagrams are to be refined in every aspect for this report (as per requirements finalized in phase I)
· Proper notations are to be used in all the figures drawn
· Proper Color-coding if required is to be used
· Extensions to diagrams / customizations may be done and represented (if the project demands it)

ELSE (groups following Structured Approach)

· The group is to draw the DFD-s (Data Flow Diagrams) for the Project. (These should be justifiable with respect to Phase I, II and the ERD)

· DFD Level 0, Level 1, Level 2 should be drawn in an evolutionary fashion (No entries to appear in Level 2 unless they are in Level 1, which in turn are in Level 0)

· The DFD-s are to be validated and made final in a presentable fashion.

· Proper notations are to be used in all the figures drawn

· Proper Color-coding is expected

· Extensions to DFD-s may be represented (if the project demands it)

V) Coding

. Algorithms/ Flowcharts

· Software used

· Hardware specification

· Programming language

· Platform

· Components

· Tools

. Coding Style Format

V) Test data Sets, Result and Analysis

VI) Testing

· Format technical reviews

· Test plan

· Test cases

· Test results

(Unit, integration, regression, system,α,ß)

VII) Configuration Management Plan

VIII) Software Quality Assurance Plan

* Costing (Time, Money and Resources)

(Do not include costing in the project report; submit to the guide)

Conclusion

Conclusions in atleast 50 words based on work done

References

List out Books, Magazines, Thesis, Journals, Web links etc referred in IEEE format

Glossary

In Keyword Alphabetical Order Ascending along with Page numbers

Plagiarism Check Report

Annexure V: Project Report Formatting Guidelines

1) **Report Size:** Limit your Project report to preferably 25- 40 pages for partial project report. Limit your Project report to preferably 80-100 pages for final project report.
2) **Footer:** The footer "Department of Computer Engineering,DIT Pune" should be included. It should be TIMES NEW ROMAN 10 pt and centrally justified.
3) **Header:** Project Title centered and page nos. on right should be included. **Start numbering from introduction.**
4) **Paper Size:** A4 Size, bond paper.
5) **Margins: Mirrored.**
    1. **Top : 1 inch**
    2. **Bottom : 1 inch**
    3. **Inside :1.25 inch**
    4. **Outside :1 inch**
6) **Line Spacing:** 1. 5 lines
7) **Title of Chapter:**
    1. **Font : Arial (Bold face, Capital,)**
    2. **Size :16 pt, Alignment:** centered
8) **All Topics heading:**
    - **i.** First order Heading: (for example -**1. Introduction**)
        1. Font : Times New Roman(Bold Face)
        2. Size : 14 pt
    - **ii.** Second order Heading: (for example -**1.1 Evolution)**
        1. Font : Times New Roman(Bold Face)
        2. Size : 12 pt
    - **iii.** Third order Heading: for example -1.1.1 Image Processing
        1. Font : Times New Roman(Normal Face)
        2. Size : 12 pt
        3.
9) **Text:**

    1. Font : Times New Roman(Bold Face)
    2. Size : 12 pt

10) **Figures and Tables:**
    1.Caption:(for figures below the figure and for tables above the table)
    2. Font: Garamond(Bold)
    3. Size:11 pt
    4. Alignment: Center
11) **References :**

. Book

Author name(s), Book Title, Publisher, Copyright Year, page nos. if any.

. Journal/ Magazine/ Periodical

Author name(s), paper name, Journal/ Magazine/ Periodical name, issue no., page nos.

.Web Resources

Complete URL including File name.

Plagiarism Check Report

Annexure VI: Format for Project Report Cover page/ Title page

(Partial) Project Report
On

Title of Project

by

Name of Student 1 (Exam NO:)
Name of Student 2 (Exam NO:)

Under the guidance of

Name of the Guide



Department of Computer Engineering

Dr.D.Y.Patil Institute of Technology
Pune-18

SAVITRIBAI PHULE PUNE UNIVERSITY
2026-27

Format for Project Approval sheet

PROJECT APPROVAL SHEET

A

Project

on

(Project Title)

Is successfully completed by

Student names (Exam NO:)

at

Department of Computer Engineering
Dr.D.Y.Patil Institute of Technology Pune-18
Savitribai Phule Pune University
20 -20

Name
Project Guide
Department of Computer Engg.

Name of Head
Head

Annexure VII: Report Documentation

Report Documentation

Report Code: CS-BE-Project 20 -20	**Report Number: < >**

Report Title:

Address (Details):
Dr.D.Y.Patil Institute of Technology Pune-
Pin – 411018, M.S. INDIA.

Author 1 [with Address, phone, E-mail]: **Address** **E-mail :** **Roll:** <Roll Number> > **Cell No**	**Author 2 [with Address, phone, E-mail]:** **Address** **E-mail :** **Roll:** <Roll Number> **Cell No**	**Author 3 [with Address, phone, E-mail]:** **Address** **E-mail :** **Roll:** <Roll Number> **Cell No**	**Author 4 [with Address, phone, E-mail]:** **Address** **E-mail :** **Roll:** <Roll Number> **Cell No**

Year: 20 – 20
~~Branch:~~ ~~Computer Engineering~~

***Key Words*:** <Keywords in the Report>

Type of Report: FINAL	Report Checked By:	Report Checked Date:	**Guides Complete Name:** <Guide's Complete Name>	Total Copies N+2

Abstract: <A Brief Abstract of the Seminar>

NOTE –
This table should not go beyond this page.
Scale down the Abstract if it does not fit in one page.
Take guide's Signature in the **"Report Checked By**:" Cell and Date of Signature in the "**Report Checked Date:**" Cell.
This page is the last page of the projects report and is NOT to be included in the **"Page Count"**

Annexure VIII: Software Engineering Code of Ethics and Professional Practices

(Courtesy / Reference- http://www.acm.org/about/code-of-ethics)

Computers have a central and growing role in commerce, industry, government, medicine, education, entertainment and society at large. Software engineers are those who contribute by direct participation or by teaching, to the analysis, specification, design, development, certification, maintenance and testing of software systems. Because of their roles in developing software systems, software engineers have significant opportunities to do good or cause harm, to enable others to do good or cause harm, or to influence others to do good or cause harm. To ensure, as much as possible, that their efforts will be used for good, software engineers must commit themselves to making software engineering a beneficial and respected profession. In accordance with that commitment, software engineers shall adhere to the following Code of Ethics and Professional Practice.

The Code contains eight Principles related to the behavior of and decisions made by professional software engineers, including practitioners, educators, managers, supervisors and policy makers, as well as trainees and students of the profession. The Principles identify the ethically responsible relationships in which individuals, groups, and organizations participate and the primary obligations within these relationships. The Clauses of each Principle are illustrations of some of the obligations included in these relationships. These obligations are founded in the software engineer's humanity, in special care owed to people affected by the work of software engineers, and the unique elements of the practice of software engineering. The Code prescribes these as obligations of anyone claiming to be or aspiring to be a software engineer.

Software engineers shall commit themselves to making the analysis, specification, design, development, testing and maintenance of software a beneficial and respected profession. In accordance with their commitment to the health, safety and welfare of the public, software engineers shall adhere to the following Eight Principles:

1. PUBLIC - Software engineers shall act consistently with the public interest.
2. CLIENT AND EMPLOYER - Software engineers shall act in a manner that is in the best interests of their client and employer consistent with the public interest.
3. PRODUCT - Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.
4. JUDGMENT - Software engineers shall maintain integrity and independence in their professional judgment.
5. MANAGEMENT - Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.
6. PROFESSION - Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.
7. COLLEAGUES - Software engineers shall be fair to and supportive of their colleagues.
8. SELF - Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.

Environment and Computing-

Information and communication technologies (ICTs) have been contributing to environmental problems: computers, electronic devices and ICT infrastructure consume significant amounts of electricity, placing a heavy burden on our electric grids and contributing to greenhouse gas emissions. In 2007, the total footprint of the ICT sector –

including personal computers (PCs) and peripherals, telecoms networks and devices and data centers – was 830 Mt CO2 emission, about 2% of the estimated total emissions from human activity released that year (a figure equivalent to aviation). ICT hardware poses severe environmental problems both during its production and its disposal. Each stage of a computer's life, from its production, throughout its use, and into its disposal, presents environmental problems. Manufacturing computers and their various electronic and non - electronic components consumes electricity, raw materials, chemicals, and water , and generates hazardous waste. All these directly or indirectly increase carbon dioxide emissions and impact the environment and the trend is to increase in the BAU (Business As Usual) scenario.

Green Computing-

Hence you all our students are requested to follow green computing practices. Green computing is the study and practice of designing, manufacturing, using, and disposing of computers, servers, and associated subsystems — such as monitors, printers, storage devices, and networking and communications systems —efficiently and effectively with minimal or no impact on the environment. Green computing includes the dimensions of environmental sustainability, the economics of energy efficiency, and the total cost of ownership, which includes the cost of disposal and recycling. Green computing benefits the environment by improving energy efficiency, lowering greenhouse gas emissions, using less harmful materials, and encouraging reuse and recycling . Green design, Green manufacturing, Green use, Green disposal are complementary paths of green ICT. Only focusing on these four fronts we can achieve total environmental sustainability from the IT side and make IT greener throughout its entire lifecycle.

Social Life and Computing-

Each IT professional must keep in mind the three key components of a corporate Green IT best practices policy -Environment, Economy and Social aspect. The invention of the computer has completely changed the way we live our lives. Nearly everything is controlled by a computer; cars, satellites, phones, etc. Computers have made our lives easier. Computers can also have positive effects on a person's social life when their power to connect over great distances is harnessed fully. Computers have both positive and negative impact in our society. While technology is a wonderful thing it is almost likely that it can be used in an immoral or wrong way. There is a price to pay for everything even if it appears it's making life easier on people.

While proper lifecycle management can greatly boost a IT company's ecological and environmental sustainability position, it can also contribute to achieving goals on the social front. Hardware retirement practices are the primary concern in this regard. In addition to seeking carbon neutrality, a proper asset retirement strategy should seek sustainability in the communities where companies operate.

The following social objectives should be considered:

1. To optimize sustainability in their IT infrastructure, companies should focus on each state of the IT lifecycle
2. Setting the Appropriate Corporate Sustainability Policy
3. Avoiding unethical labor practices and Controlling unethical exports
4. Accountability in the Recycling e-waste and Sustainability Metrics and Reporting
5. Greater Transparency Regarding Material Analysis and Extraction
6. Compliance with stringent, evolving security regulations

Annexure IX: Copy of Project Proposal

Part A follows the format of Annexure I (Final Proposal) and Part B the format of Annexure II (System Requirement Specification). Fields marked "to be filled" are left for the team or the guide.

Part A: Project Proposal

A.1 Title Page

Field	Details
Project Group ID	_____ (to be filled)
Title of the project	Cortex: An Agentic Retrieval-Augmented Generation System over Live Work Data with Explainable Memory and Consent-Gated Actions
Domain	Information Retrieval and Natural Language Processing (Agentic RAG); Web Technology
Sponsorship details	Not sponsored (internal project); no external guide
Internal Guide	Prof. Swapnali Gaikwad (swapnali.gaikwad@dypvp.edu.in); Designation: _____ (to be filled); Signature of approval: _____
Type of Project	Non-sponsored; System: application software (full-stack web application with an AI agent)
Mentor (PG student / alumni / other)	_____ (to be filled)
TE students associated (Roll No, Name)	_____ (to be filled)

Team Members

PRN	Roll No	Name	Role / Task assigned	E-mail	Mobile	Sign
	BCOC17	Ajit Paraskar		ajitparaskar87@gmail.com		
	BCOC32	Harshvardhan Patil		harsh.patil.tech@gmail.com		
	BCOC35	Prathamesh Kumbhar		prathameshkumbhar.dev@gmail.com		
	BCOC47	Bikram Ghosh		ghoshbikram2725@gmail.com		

PRN, role, mobile number and signature are left blank on purpose.

A.2 Keywords (ACM Computing Classification System, 1998)

- H.3.3 Information Search and Retrieval: query formulation, retrieval models, search process, information filtering
- H.3.4 Systems and Software: question-answering (fact retrieval) systems, distributed systems, performance evaluation
- I.2.7 Natural Language Processing: language generation, language models
- I.2.8 Problem Solving, Control Methods, and Search: plan execution, formation, and generation
- I.2.11 Distributed Artificial Intelligence: intelligent agents

Free keywords: agentic RAG, query decomposition, claim verification, groundedness, explainable memory, human-in-the-loop consent, Model Context Protocol, free-tier LLM failover.

A.3 Problem Definition

To design and implement an agentic retrieval-augmented generation system that answers natural-language questions over a user's live, distributed work data (e-mail, calendar, documents, notes, team chat, code hosting and the web) by decomposing each question into sub-questions, retrieving evidence from all relevant sources in parallel within fixed budgets, and producing an answer in which every factual claim is cited and verified against its evidence; that surfaces contradictions between

sources instead of hiding them, says plainly when information is missing, keeps an explainable and user-governed long-term memory, and performs actions in the user's tools only after explicit, per-action consent; all while running on free-tier language model services.

A.4 Abstract

Work information is scattered across Gmail, Google Calendar, Google Drive, Notion, Slack, GitHub and the web, and it changes every day. Ordinary retrieval-augmented chatbots run one similarity search over a static corpus, treat every sentence as equally certain, resolve contradictions silently and either cannot act or act without asking. Cortex is an agentic retrieval-augmented generation system that plans two to five sub-questions, searches the connected sources in parallel through a registry of interchangeable connectors, recalls remembered priorities, and streams an answer in which every factual sentence cites numbered evidence. A verifier checks each claim lexically (the citation exists, the content words overlap and the numbers match) and, in Deep mode, asks a model to re-check only the doubtful claims; the result is a groundedness score and a confidence state for each claim. Contradictions are reported with both citations, missing information is stated without spending a model call, and connector failures are explained together with their impact on the answer. Actions are proposed as confirmation cards and executed only through a single approval endpoint. A Python memory service keeps provenance, salience decay, expiry, pinning, editing and forgetting for every memory. A failover router over free-tier providers with cooldowns and a response cache keeps the operating cost at zero. A preliminary 12-question evaluation gave a mean groundedness of 0.88 and a citation recall of 0.92.

A.5 Process Diagram, System Architecture, Modules and Functionalities

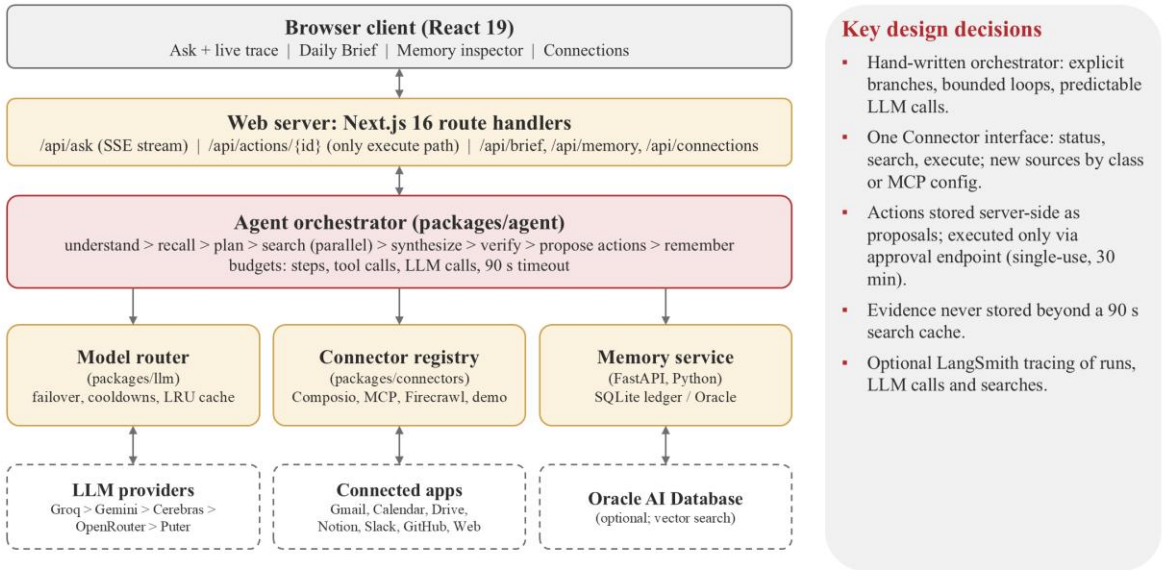


Figure 1: System architecture of Cortex

Process (one agent run): understand (check which sources are available) → recall memories → plan 2-5 sub-questions → search all (sub-question × source) pairs in parallel → synthesize a streamed, cited answer → verify each claim → propose actions (never execute) → remember salient facts.

Module	Functionality
Web application (apps/web)	Ask screen with live trace, Daily Brief, Memory inspector, Connections; REST and server-sent event API
Agent orchestrator (packages/agent)	Bounded pipeline, prompts, claim verifier, action store, Daily Brief, tracing
Model router (packages/llm)	OpenAI-compatible failover over Groq, Gemini, Cerebras, OpenRouter and Puter; cooldowns; response cache

Connectors (packages/connectors)	Uniform search/execute interface over Composio (six apps), any MCP server, Firecrawl (web) and a sample workspace
Memory service (services/memory)	FastAPI service with SQLite ledger and optional Oracle Agent Memory; provenance, decay, expiry, pin, edit, forget
Shared types (packages/shared)	Zod schemas and types used by server and client
Evaluation harness (evals)	12-question evaluation over the sample workspace; groundedness, citation recall, honesty, consent safety

A.6 Literature Survey

Work	Contribution	Limitation for this problem
Lewis et al., RAG (NeurIPS 2020)	Generator conditioned on passages from a dense index	One retrieval round over a fixed index
Yao et al., ReAct (ICLR 2023)	Interleaves reasoning with tool actions	Open-ended loop; no claim-level check
Trivedi et al., IRCoT (ACL 2023)	Retrieval interleaved with chain-of-thought	Static corpus; many model calls
Press et al., Self-Ask (EMNLP 2023)	Explicit follow-up sub-questions	No source governance or verification
Asai et al., Self-RAG (ICLR 2024)	On-demand retrieval with self-critique	Requires fine-tuning
Yan et al., CRAG (2024)	Retrieval evaluator with web fallback	Trained evaluator; web only
Jeong et al., Adaptive-RAG (NAACL 2024)	Routes questions by complexity	Trained classifier; no live tools
Gandhi et al., GRASP (2026), base paper	RL policy over semantic search, keyword search and paragraph reading	Needs RL training; benchmark corpora only
Gao et al., ALCE; Min et al., FActScore (EMNLP 2023)	Metrics for citation quality and factual precision	Offline evaluation, not a system
Packer et al., MemGPT (2023); Park et al., Generative Agents (UIST 2023)	Long-term agent memory by relevance, recency, importance	Memory not visible or controllable by the user
Hou et al., MCP survey (2025)	Ecosystem and security threats of the Model Context Protocol	Highlights risk of unconsented tool actions

Current market survey (similar systems, as surveyed by the team on 25 October 2026):

System	Pros	Cons / gap filled by Cortex
Supermemory	Memory infrastructure and API with plug-ins for assistants	Memory back-end only; no planning agent or cited multi-hop research over live sources
MemoryFirst	Graph memory, citations, morning brief	Resolves contradictions silently (latest wins); paid beyond one project
Augmas	Enterprise search and chat across many tools with citations	No cross-session memory, no visible reasoning; enterprise pricing
OpenAI Dots	Always-on agents with memory	Closed, paid, opaque memory
Hermes Agent (Nous Research)	Personal agent with file-based memory	Flat-file memory; no evidence graph or project-analysis connectors

A.7 Objectives

- Answer multi-part questions over several live sources with a bounded plan of sub-questions and parallel retrieval.
- Cite every factual sentence and verify each claim against the evidence it cites; report a groundedness score.
- Surface cross-source contradictions as risks with both citations, and state missing information instead of guessing.

- Never execute an action in a user's tool without an explicit, single-use approval of an exact preview.
- Keep long-term memory selective and explainable (reason, source, salience, expiry), with pin, edit and forget controls.
- Degrade gracefully: name any source that fails, explain the impact, and continue with the rest.
- Keep the number of model calls per question small and predictable (at most four in Deep mode, two in Fast mode).
- Provide an evaluation harness and an automated test suite that measure groundedness, citation recall, honesty and consent safety.

A.8 Scope of the Project

- **In scope:** a web application and reusable packages (agent, connectors, model router) plus a memory service that answer questions over Gmail, Google Calendar, Google Drive, Notion, Slack, GitHub, any MCP server and the web, with cited, verified answers, action proposals and an explainable memory.
- **Bounds:** question <= 2000 characters; history <= 6 turns; plan <= 5 sub-questions; <= 18 search calls per run; <= 16 evidence items (Deep) or 8 (Fast); <= 3 action proposals and <= 2 memory writes per run; 90 s run timeout.
- **Out of scope:** copying users' data into an index or warehouse (tools are queried at question time, with a 90 s cache), training or fine-tuning models, autonomous write operations, and the vision items deferred to future work (time-travel queries, editable plan, Cortex as an MCP server).

A.9 Software and Hardware Requirements

Hardware parameter	Minimum requirement	Justification
CPU	2 GHz, dual core	Runs the Next.js server, memory service and browser; inference happens at the providers
RAM	8 GB	Node.js, Python service and browser together; more for the optional Oracle container
Storage	5 GB free	Dependencies, virtual environment, caches and the SQLite ledger
Network	Broadband Internet	Every model call and live connector call is a remote API request

- **Operating system:** Windows 10/11, Linux or macOS.
- **Languages:** TypeScript on Node.js 24 or later; Python 3.11.
- **Frameworks and libraries:** Next.js 16, React 19, Tailwind CSS 4, Zod, Composio SDK, MCP TypeScript SDK, Firecrawl SDK, LangSmith SDK, FastAPI, Uvicorn, Pydantic, Vitest; oracledb and oracleagentmemory (optional).
- **External services (free tiers):** Groq, Google Gemini, Cerebras, OpenRouter and Puter (models); Composio; Firecrawl; LangSmith; Oracle AI Database (optional).
- **Tools:** Visual Studio Code, Git.

A.10 Expected Outcomes

- A working web application (Ask with live trace, Daily Brief, Memory, Connections) that answers questions over six work tools and the web with cited, verified answers.
- Reusable packages: a free-tier model router with failover, cooldowns and caching; a connector registry; and a memory service with provenance, decay and governance.
- A consent model in which only one endpoint can execute an action, once, for the user who received the proposal, within 30 minutes.

- An automated test suite and an evaluation harness. Preliminary results recorded so far: mean groundedness 0.88, citation recall 0.92, zero actions executed without consent (12 questions, synthetic workspace).
- A review paper on agentic RAG with Cortex as a case study (manuscript prepared, not yet submitted).

A.11 Probable Date of Completion

As per the departmental schedule: submission of the final project report in the first week of April 2027 (tentative; to be confirmed with the guide).

A.12 References

[1] V. Gandhi, J. Lee, S. Todmal, F. Dernoncourt, R. Rossi, Z. Wang, and A. Lan, "GRASP: GRanularity-aware search policy for agentic RAG," arXiv:2607.10463, 2026.

[2] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in Proc. NeurIPS, vol. 33, 2020, pp. 9459-9474.

[3] S. Yao et al., "ReAct: Synergizing reasoning and acting in language models," in Proc. ICLR, 2023.

[4] H. Trivedi, N. Balasubramanian, T. Khot, and A. Sabharwal, "Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions," in Proc. ACL, 2023, pp. 10014-10037.

[5] O. Press, M. Zhang, S. Min, L. Schmidt, N. A. Smith, and M. Lewis, "Measuring and narrowing the compositionality gap in language models," in Findings of EMNLP, 2023, pp. 5687-5711.

[6] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-RAG: Learning to retrieve, generate, and critique through self-reflection," in Proc. ICLR, 2024.

[7] S.-Q. Yan, J.-C. Gu, Y. Zhu, and Z.-H. Ling, "Corrective retrieval augmented generation," arXiv:2401.15884, 2024.

[8] S. Jeong, J. Baek, S. Cho, S. J. Hwang, and J. C. Park, "Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity," in Proc. NAACL, 2024.

[9] B. Jin et al., "Search-R1: Training LLMs to reason and leverage search engines with reinforcement learning," arXiv:2503.09516, 2025.

[10] T. Gao, H. Yen, J. Yu, and D. Chen, "Enabling large language models to generate text with citations," in Proc. EMNLP, 2023, pp. 6465-6488.

[11] L. Gao et al., "RARR: Researching and revising what language models say, using language models," in Proc. ACL, 2023, pp. 16477-16508.

[12] S. Min et al., "FActScore: Fine-grained atomic evaluation of factual precision in long form text generation," in Proc. EMNLP, 2023, pp. 12076-12100.

[13] S. Es, J. James, L. Espinosa Anke, and S. Schockaert, "RAGAs: Automated evaluation of retrieval augmented generation," in Proc. EACL: System Demonstrations, 2024, pp. 150-158.

[14] C. Packer, V. Fang, S. G. Patil, K. Lin, S. Wooders, and J. E. Gonzalez, "MemGPT: Towards LLMs as operating systems," arXiv:2310.08560, 2023.

[15] J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," in Proc. ACM UIST, 2023.

[16] T. Schick et al., "Toolformer: Language models can teach themselves to use tools," in Proc. NeurIPS, vol. 36, 2023.

[17] X. Hou, Y. Zhao, S. Wang, and H. Wang, "Model Context Protocol (MCP): Landscape, security threats, and future research directions," arXiv:2503.23278, 2025.

[18] Anthropic, "Introducing the Model Context Protocol," Nov. 2024. [Online]. Available: https://www.anthropic.com/news/model-context-protocol

[19] A. Singh, A. Ehtesham, S. Kumar, and T. T. Khoei, "Agentic retrieval-augmented generation: A survey on agentic RAG," arXiv:2501.09136, 2025.

[20] N. F. Liu et al., "Lost in the middle: How language models use long contexts," Trans. ACL, vol. 12, pp. 157-173, 2024.

[21] M. Chui et al., "The social economy: Unlocking value and productivity through social technologies," McKinsey Global Institute, 2012.

[22] M. Chui et al., "The economic potential of generative AI: The next productivity frontier," McKinsey & Company, Jun. 2023.

[23] R. S. Pressman and B. R. Maxim, Software Engineering: A Practitioner's Approach, 8th ed. New York, NY, USA: McGraw-Hill Education, 2015.

Part B: System Requirement Specification

B.1 Detailed Problem Definition

A user's work data lives in several independently hosted tools. Given a question q from this user, Cortex must, within a bounded number of model calls and tool calls, produce an answer that (a) uses only evidence retrieved from those tools during the run, (b) cites the evidence for every factual sentence, (c) reports how much of the answer is supported (groundedness), (d) reports where the sources disagree and what could not be found, and (e) turns any requested write operation into a proposal that the user must approve before it runs. The system must keep working when individual sources or model providers fail, and it must remember only what is likely to matter later, in a way the user can inspect and control.

Necessary functions

- Ask a question in Fast or Deep mode and receive a streamed answer with numbered citations.
- Verify every claim and show its state (supported, partial, unsupported) and the groundedness score.
- Report conflicts between sources, missing information and failed sources with their impact.
- Propose actions as confirmation cards; execute only after approval through one endpoint.
- Connect the six work applications through managed OAuth, or use the sample workspace.

Desirable functions

- Daily Brief of what changed, what is at risk, conflicts and what is upcoming.
- Memory inspector to teach, pin, edit or forget memories, with reason, source and age.
- Adding any MCP server through a configuration file; optional LangSmith tracing.

Others (future work)

- Time-travel queries over source snapshots, an editable plan before execution, shareable reasoning receipts, Cortex as an MCP server, and a hosted deployment with persistent pending actions.

B.2 External Interface Requirements

User interfaces

- **Ask:** chat screen that streams the answer; citation chips open the quoted evidence and a link to the source; unsupported claims are struck through; a collapsible trace shows the plan, each tool call with its result count and latency, and model usage.
- **Action card:** exact preview of a proposed action with Edit, Approve and Decline.
- **Daily Brief:** cited cards grouped as Changed, At risk, Conflicts and Upcoming, each with “Ask about this”.
- **Memory:** list of memories with source, reason, salience and age, and Pin, Edit and Forget controls.
- **Connections:** grid of six application tiles with Connect / Connected status, and a sample-workspace switch.
- The interface is responsive (laptop and phone), uses plain-language error messages, and needs no settings pages or tokens from the end user.

Hardware interfaces

- No special hardware is required. The system runs on a standard computer (2 GHz dual-core CPU, 8 GB RAM, 5 GB free storage) with a network interface; all model inference takes place at remote providers.

Software interfaces

Interface	Purpose
POST /api/ask	Run the agent; the response is a server-sent event stream of typed events
POST /api/actions/{id}	Approve or reject one proposal; the only execution path
GET /api/brief	Daily Brief
GET, POST /api/memory; PATCH, DELETE /api/memory/{id}	List and teach memories; pin, edit, forget
GET, PUT /api/connections; POST /api/connections/{id}	Connection status, workspace choice, start OAuth
Memory service: /health, /memories, /search, /threads	Status and backend, memory CRUD, recall, conversation threads
OpenAI-compatible chat completions	Groq, Gemini, Cerebras, OpenRouter, Puter
Composio SDK / hosted MCP	Gmail, Calendar, Drive, Notion, Slack and GitHub tools and OAuth
MCP (streamable HTTP or stdio)	Any server declared in mcp.config.json
Firecrawl API; LangSmith API	Web search and scraping; optional tracing

Communication interfaces

- HTTPS between the browser and the web server; answers are streamed as server-sent events (text/event-stream).
- HTTP/JSON between the web server and the memory service on the local machine (port 8000); the web server listens on port 3000.
- HTTPS REST calls to model providers, Composio, Firecrawl and LangSmith; MCP over streamable HTTP or standard input/output.
- OAuth 2.0 authorisation for connected applications is handled by Composio; tokens never reach the browser.
- The anonymous user identifier is carried in an HTTP-only, same-site cookie.

B.3 System Features

Feature 1: Ask with cited, verified answers

The user submits a question (<= 2000 characters) and a mode. The agent plans sub-questions, searches sources in parallel, streams an answer that may use only numbered evidence, and verifies each claim. Response: answer, claim states, groundedness, conflicts, missing items and follow-up questions.

Feature 2: Fast and Deep modes

Fast mode uses a local single-step plan and one model call; Deep mode uses a model plan, an empty-result retry and a model re-check of doubtful claims (at most four model calls).

Feature 3: Trace and evidence inspection

Each run streams step, plan, tool-call, evidence and usage events; the user can open any citation and the full trace.

Feature 4: Consent-gated actions

Model-drafted actions are validated against the connector's declared actions and parameters, stored as pending proposals with an exact preview, and executed only after approval; single-use, scoped to the user, expiring after 30 minutes.

Feature 5: Daily Brief

Proactive cited cards for what changed recently and what is due soon, ranked by remembered priorities.

Feature 6: Explainable memory

Explicit “remember” requests (salience 0.95) and model candidates with salience >= 0.6 (at most two per run) are stored with reason, source and expiry; salience decays with a 30-day half-life unless pinned; the user can pin, edit or forget.

Feature 7: Connections and sample workspace

One-click OAuth through Composio for six applications; a fictional sample workspace with deliberate cross-source conflicts allows demonstration and evaluation without personal accounts.

Feature 8: Free-tier model router

Failover across providers with cooldowns (429: retry-after or 60 s; bad key: 10 min; out of credit: 1 h) and a 6-hour response cache.

B.4 Other Non-functional Requirements

Performance requirements

- The answer should begin streaming within a few seconds; searches run concurrently.
- Timeouts: 12 s per connector call (one retry), 6 s per connector status check, 25 s per model request, 5 s per memory request, 90 s per run.
- Budgets per run: <= 5 sub-questions, <= 18 tool calls, <= 4 model calls in Deep mode and <= 2 in Fast mode.
- Recorded so far (preliminary, Deep mode, 12 questions): median end-to-end time 3.7 s and 90th percentile 7.1 s.

Safety requirements

- No write operation is executed without explicit, single-use approval of an exact preview; a second decision is refused (HTTP 409).
- Users' source data is not stored beyond short-lived caches (90 s search cache); OAuth tokens are never sent to the browser.
- If a source, model provider or the memory service fails, the run continues, and the failure and its impact are reported to the user.
- If no evidence is found, the system says so instead of guessing.
- Memories of the sample workspace are kept separate from real ones.

Software quality attributes

- **Availability:** provider failover with cooldowns; failed sources skipped and reported; SQLite fallback when Oracle is not available.
- **Modifiability:** a new source is one connector class or one MCP configuration entry; tool names and model identifiers are data.
- **Portability:** runs on Windows, Linux and macOS.
- **Security:** single approval endpoint; HTTP-only, same-site cookie; tokens held by Composio.
- **Testability:** all external dependencies are injected and can be replaced by stubs; 18 automated tests and an evaluation harness.
- **Usability:** plain-language messages; claims visually marked by verification state; one-click connection.